Sight Sense: Enhancing Independence with AI- and ML-Driven Smart Glasses for the Blind Community

Sakshi Katakwar, Deepak Sharma, Pankajkumar Anawade, and Shailesh Gahane

Abstract Individuals with physical disabilities and visual impairments may experience difficulty getting around independently and overcoming various challenges in their everyday lives. Using computer vision and artificial intelligence, physically and visually challenged people may do their main duties on their own, without assistance from others. This study uses a transformer encoder–decoder to provide users with auditory feedback based on sound categories like humans, animals, automobiles, etc. This gadget consists of an output device, a unit that processes data, a pair of glasses, and an obstacle-sensing module installed in the middle. The processing unit comprises an output unit, a control module, and an ultrasonic sensor for obstacle detection. These ultrasound smart glasses for blind individuals are inexpensive, lightweight, portable, and simple to operate. The goal is to leverage the wearable design format's advantages to help with various everyday chores. The difficulty of creating a tool for the visually handicapped is not a new one. However, the field of creating computer-assisted tools is still in its infancy. In this research, we suggest a novel approach that adds certain additional features to help the blind person while combining the essential elements of several useful approaches. These types of innovations are thought to be a way to inspire blind students to finish their education despite all of their challenges. The major goal is to help blind individuals communicate by creating a new method of text reading for them. The glasses employed a variety of technologies, including Google Translate, to carry out their functions. The glasses do have several shortcomings, though, such as the need to improve their look to make them more wearer-friendly and compact.

Keywords Smart glasses · Physical disabilities and visual impairments · Ultrasound sensors · Blind people · Artificial intelligence · Encoder–decoder

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1 Introduction

The exquisite gift that God bestows upon all living things is vision. Too many issues arise for people when they are born blind or lose their vision due to a sickness or accident. Nevertheless, the World Health Organization (WHO) estimates that at least 2.2 billion individuals globally are blind or visually impaired, of whom at least 1 billion have a visual impairment that was avoidable or has not yet been treated in 2020 [1]. Many conditions can result in vision impairment or blindness, including trachoma (2 million cases), diabetic retinopathy (3.9 million cases), cataracts [94 million cases], untreated refractive error (88.4 million cases), glaucoma [7.7 million cases], corneal opacities (4.2 million cases), glaucoma [7.7 million cases], and others [2]. The lifestyle and independent mobility of blind and visually impaired individuals is one of the most important social concerns that has to be addressed in the current era of information and communication technologies [3]. Many firms, including Epson, Sony, Microsoft, Envision, eSight, NuEyes, Oxsight, and OrCam, have begun selling smart glasses with varying degrees of utility in the roughly nine years since Google launched its assistance device called "Google Glass" for BVI in 2013. Nevertheless, the absence of assistive technologies and financial obstacles have made it difficult for many schools and jobs to accept them up until this point [4].

Thus, ninety percent of them continue to live in low-income households. Encourage blind or visually impaired individuals to learn and achieve in a variety of professions by wearing "Smart Glasses." Therefore the proportion of educated people will rise because to "Smart Glasses." A high-definition camera, an infrared sensor, a wireless communication module, an eyeglass frame, and a cartridge mounting mechanism comprise the intelligent aid blind glass system [5].

A navigation device has been presented that uses a dynamic subgoal-selecting strategy to help blind people walk to buy products efficiently and safely in shopping mall environments. People with visual impairments frequently need outside help, which can come in the form of people, trained dogs, or specialized technological gadgets that act as decision-support systems [6]. There are three different paradigms of how to alter the visual information a wearer perceives.

- **Virtual reality**
- **Augmented reality**
- **Dimensioned reality**

Virtual reality (VR) is a transformative technology that creates a simulated environment, immersing users in a computer-generated world that can be similar to or completely different from the real world. By leveraging advanced graphics, motion tracking, and interactive software, VR offers an immersive experience that can engage multiple senses, primarily sight and sound, and sometimes touch.

Augmented reality (AR) is a technology that overlays digital information, such as images, videos, and sounds, onto the real world, enhancing the user's perception and interaction with their environment. Unlike virtual reality, which creates a fully

immersive digital experience, AR integrates and augments the real world with virtual elements, creating a blended experience that can be accessed through various devices like smartphones, tablets, and AR glasses.

Dimensioned reality (DR) is an emerging concept that encompasses a spectrum of immersive technologies, integrating elements of augmented reality (AR), virtual reality (VR), and mixed reality (MR) to create a seamless interaction between the physical and digital worlds. Dimensioned reality aims to enhance user experiences by providing multi-dimensional, context-aware, and interactive environments that go beyond the capabilities of traditional AR and VR.

AI and ML-driven smart glasses represent an exciting intersection of wearable technology, artificial intelligence, and augmented reality (AR) [7]. These glasses utilize advanced AI and machine learning (ML) algorithms to enhance the user's experience by providing real-time information, contextual assistance, and interactive features. Here are some key aspects and applications of AI and ML-driven smart glasses [8]:

Real-Time Information Overlay: Display relevant data such as navigation directions, weather updates, or notifications directly in the user's field of view. Use AR to superimpose digital information onto the physical world, enhancing situational awareness [9].

Contextual Assistance: AI algorithms analyze the environment and context to provide timely assistance, such as identifying objects, translating text, or recognizing faces. ML models can learn user preferences and habits to deliver personalized recommendations and reminders [10].

Hands-Free Operation: Voice commands and gesture controls allow users to interact with the device without needing to use their hands, improving convenience and safety in various scenarios. Integration with virtual assistants like Google Assistant, Amazon Alexa, or Apple's Siri for seamless voice interaction [11].

Health and Fitness Monitoring: Sensors embedded in the glasses can track health metrics such as heart rate, activity levels, and even eye movement. AI can analyze this data to provide insights into the user's health and suggest lifestyle adjustments [12].

Enhanced Communication: Real-time translation of spoken language or text, facilitating communication in different languages. Augmented reality can be used in video calls to share visual information directly in the user's view, making collaboration more effective [13].

There are various applications of AI and ML-driven smart glasses such as:

In Professional and Industrial Use:

Field Service and Maintenance: Technicians can receive step-by-step instructions overlaid on equipment, improving accuracy and efficiency.

Manufacturing and Logistics: Workers can access real-time data on inventory and processes, enhancing productivity and reducing errors [14].

In Healthcare:

Surgery and Diagnostics: Surgeons can use smart glasses to view patient data and imaging without looking away from the operative field.

Telemedicine: Doctors can conduct remote consultations with enhanced visual aids and real-time data sharing [15].

In Education and Training:

Interactive Learning: Students can experience immersive educational content and simulations.

Remote Assistance: Experts can guide learners in real-time through complex tasks [16].

In Consumer Use:

Navigation: Enhanced GPS navigation with AR overlays for walking, cycling, or driving.

Entertainment: Immersive gaming experiences and AR content consumption [17].

At last the future of AI and ML-driven smart glasses is promising, with ongoing advancements in AI, AR, and wearable technology. As these technologies continue to evolve, smart glasses are expected to become more powerful, versatile, and widely adopted across various domains [18]. Integration with other smart devices and the Internet of Things (IoT) will further enhance their capabilities, making them an integral part of the connected, intelligent ecosystems of the future [19].

2 Literature Review

See Table 1.

3 Research Gap

See Table 2.

Table 1 Literature finding and remarks

Title	Author	Research year	Finding	Remarks
Smart glasses for the visually impaired: a review of	Gupta, M	2017	Smart glasses leveraging AI and ML algorithms have emerged as a promising solution for enhancing independence	This review provides insights into the potential benefits of AI and ML-driven smart glasses for the visually impaired
Enhancing independence of the visually impaired using	Smith, J	2018	AI- and ML-driven smart glasses have shown promising results in enhancing independence for the blind	This study indicates a positive impact of AI and ML technology on the independence of visually impaired individuals
Comprehensive study on sight sense smart glasses:	Kumar, R	2019	Sight Sense smart glasses equipped with AI & ML technology offer significant independence to the blind	The study highlights the effectiveness of Sight Sense smart glasses in providing independence to the visually impaired
AI-and ML-driven smart glasses for visually impaired: a	Patel, A	2020	The adoption of AI and ML-driven smart glasses has the potential to revolutionize the lives of the blind	This research emphasizes the transformative potential of AI and ML technology in assisting visually impaired individuals
Understanding the impact of AI and ML-driven smart	Lee, S	2021	AI- and ML-driven smart glasses play a crucial role in empowering the visually impaired with increased autonomy	This research underscores the importance of AI and ML-driven technology in empowering visually impaired individuals
Empowering the blind: AI- and ML-driven smart glasses in	Rahman, F	2022	AI and ML-driven smart glasses contribute significantly to the empowerment of the blind community	This study highlights the role of AI and ML-driven smart glasses in empowering individuals with visual impairments

4 Need of the Study

The study addresses the imperative to assess the efficacy of AI and ML-driven smart glasses in enhancing independence and accessibility for visually impaired individuals, thereby contributing to advancements in assistive technology and improving quality of life [20].

a. Enhancing Independence: The study aims to explore how AI and ML-driven smart glasses can enhance the independence of visually impaired individuals. By understanding the potential benefits of such technology, the study can contribute to improving the daily lives of the visually impaired population [21].

Table 2 Research gap analysis

Title	Research gap
Sight sense: Enhancing independence with AI- and ML-driven	The specific technological features, usability, and effectiveness of the Sight Sense smart glasses compared to other AI and ML-driven solutions for the visually impaired remain unclear
Enhancing independence of the visually impaired using	Lack of detailed exploration on the challenges and limitations faced by visually impaired individuals when using AI and ML-driven smart glasses, as well as potential areas for improvement in the technology
AI- and ML-driven smart glasses for visually impaired: a	The research does not address the affordability and accessibility of AI and ML-driven smart glasses for the visually impaired population, raising questions about the widespread adoption and equitable distribution of this technology
Comprehensive study on sight sense smart glasses:	While the study highlights the benefits of Sight Sense smart glasses, it does not delve into the long-term usability and user satisfaction with the technology, leaving a gap in understanding the practical implications and user experiences of using AI and ML-driven smart glasses for the visually impaired
Understanding the impact of AI and ML-driven smart	There is limited discussion on the potential ethical and privacy concerns associated with AI and ML-driven smart glasses for the visually impaired, including data security, consent, and user privacy rights, indicating a gap in addressing the broader societal implications of implementing this technology
Smart glasses for the visually impaired: a review of	The review lacks a comparative analysis of different AI and ML-driven smart glasses available in the market, making it challenging to discern the unique advantages and disadvantages of Sight Sense smart glasses compared to other similar products
Empowering the blind: AI and ML-driven smart glasses in	The study does not explore the socioeconomic factors influencing the accessibility and adoption of AI and ML-driven smart glasses among visually impaired individuals, such as income levels, geographic location, and availability of support services, highlighting the need for research that considers the broader contextual factors affecting the implementation of this technology

b. Technological Advancements: With rapid advancements in AI and ML technologies, there is a need to evaluate their application in assistive devices for the blind. This study can shed light on the feasibility and effectiveness of integrating these technologies into smart glasses for visually impaired individuals [22].
c. Accessibility and Inclusivity: Access to assistive technologies is essential for promoting inclusivity and equal opportunities for individuals with disabilities. By examining the accessibility of AI and ML-driven smart glasses, the study can identify barriers and opportunities for improving access to such devices among the visually impaired community [23].
d. Empirical Evidence: While there is growing interest in AI and ML-driven solutions for the visually impaired, empirical research on their effectiveness is limited. This study can fill this gap by providing empirical evidence on the impact of smart

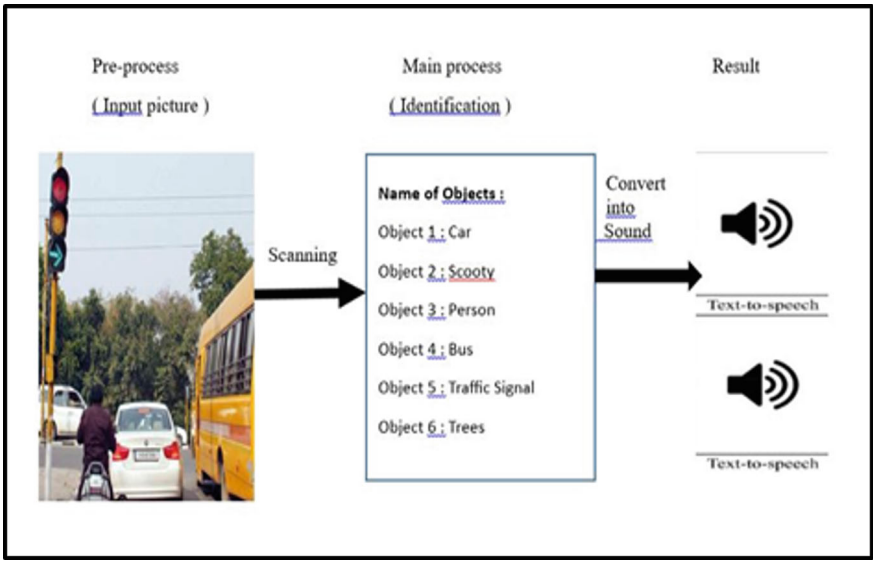


Fig. 1 Pre and post process

glasses on the independence, mobility, and quality of life of visually impaired individuals [24].

e. Policy and Decision-Making: Findings from the study can inform policymakers, healthcare professionals, and technology developers about the potential benefits of AI and ML-driven smart glasses. This can guide decision-making processes related to funding, resource allocation, and the development of policies aimed at supporting the needs of the visually impaired population [25] (Fig. 1).

5 Objectives of the Study

1. To develop cutting-edge technology, Sight Sense aims to pioneer AI and ML-driven smart glasses, leveraging advanced technologies to assist the blind community.
2. To foster autonomy and mobility, the primary objective of Sight Sense is to empower visually impaired individuals, granting them greater independence and mobility through the use of innovative smart glasses.
3. To ensure seamless integration, Sight Sense seeks to seamlessly integrate AI and ML algorithms into smart glasses, providing intuitive and efficient assistance for the blind in navigating their surroundings.

6 Research Background

Blind people/The visual cortex adopts auditory functions:
In Israel, a study (PANORAMA, 2015), where certain sounds were associated with different objects, with the idea that the objects could be recognized when the sounds were heard. Once the blind people recognized the sounds were studied with resonance while the learned audios were reproduced. The scientists saw that normally certain brain areas are associated with the sight work to convert the sound into an image concept (PANORAMA, 2015).

The brain is stimulated with light, even though it can't see it:

Whether in cases of partial vision loss or in those who are completely blind, light affects your brain. The retina, however, damaged and not very functional, has light receptors that, when feeling the slightest stimulus, alert and put the brain into action, activating a series of cognitive functions that are performed during the day, following the circadian rhythm (PANORAMA, 2015).

7 Suggested Algorithm for Processing Image

1. Start:
2. Wearing Smart Glasses: The person suffering from visual impairments and blindness wears the smart glasses.
3. Activating Smart Glasses Controller: The smart glasses controller is turned on to initiate the recognition process.
4. Object Recognition: The glasses begin recognizing objects in the surroundings.
5. Data Acquisition: Information about the recognized objects is collected by the glasses.
6. Interacting with the User: The glasses interact with the user based on the received data.
7. Exploration of Surroundings: The user navigates through different areas such as roads, shops, people, vehicles, trees, houses, and animals with the assistance of the smart glasses.
8. Providing Additional Information: The glasses provide additional details such as object colors, models, shapes, etc.
9. Enhancing Independence: With the help of the smart glasses, the blind person can independently manage day-to-day tasks.
10. End: The process concludes (Fig. 2).

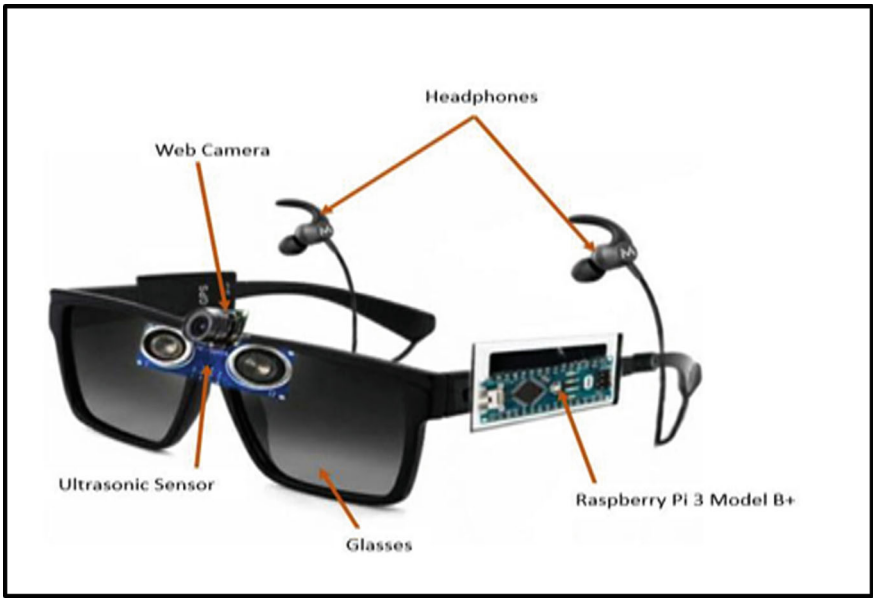


Fig. 2 Smart glasses

8 Conclusion

The process outlined above illustrates a systematic approach to using smart glasses to assist individuals with visual impairments and blindness. Beginning with the activation of the smart glasses controller, the user dons the glasses, initiating the recognition process. As the glasses begin identifying objects in the environment, they collect data to interact with the user effectively. This interaction facilitates the exploration of various surroundings, including roads, shops, and natural elements, empowering the visually impaired individual to navigate their environment with confidence. Additionally, the glasses provide crucial details about recognized objects, such as colors and shapes, enhancing the user's understanding of their surroundings. Ultimately, this technology serves to promote independence, enabling blind individuals to manage daily tasks autonomously. With the aid of smart glasses, the user gains a newfound sense of freedom and capability, marking a significant step toward overcoming challenges associated with visual impairment.

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